Keywords: bilevel reinforcement learning • inverse RL • hypergradient • black-box optimization

Reward-Estimated Hypergradient for Bilevel Reinforcement Learning with Black-Box Follower

Inverse RL with Bias Cancellation and Eigenvalue Truncation Unlocks Bilevel RL Without Access to the Follower's Reward

NeurIPS 2026 Submission
arXiv:2611.11998 • cs.LG / cs.AI • NeurIPS 2026

Bilevel RL requires the leader to differentiate through the follower's best-response, which presupposes access to the follower's reward. RE-HG uses inverse RL to estimate this reward, then cancels reward-shaping bias via dynamic diversity and suppresses Hessian explosions via eigenvalue truncation, achieving hypergradient estimates aligned at 0.91 cosine similarity with a white-box oracle.

AT A GLANCE

Problem: Bilevel RL hypergradient estimation requires the follower's reward, but in real settings the follower's reward is hidden.

Why it matters: Mechanism design, incentive alignment, and multi-agent learning all involve a leader optimizing against an unknown follower reward.

Method: Estimate the follower's reward via IRL; cancel reward-shaping bias by aggregating across diverse dynamics; truncate Hessian eigenvalues to prevent norm explosions.

Result: Hypergradient cosine similarity 0.91 vs. oracle; outperforms oracle on sharp reward landscapes (oracle gets stuck in local minima).

Evidence: Navigation, manipulation, and mechanism design benchmarks with white-box oracle comparison.

The Big Picture

leader sets parameters θ_L (a mechanism, an incentive structure, or a training curriculum), and a follower best-responds with policy $\pi_F^*(\theta_L)$ that maximizes their own reward R_F. The leader's objective depends on the follower's response, so optimizing θ_L requires the hypergradient—the gradient of V_L with respect to θ_L accounting for how changes in θ_L change π_F^*.

Computing the hypergradient involves differentiating through the follower's best-response optimality condition, which requires the follower's reward R_F (to compute the value-function Hessian). In the black-box setting, only follower trajectories are observable, not R_F. Real-world applications are predominantly black-box: a government setting tax policy against firms whose profit functions are private; an online platform designing recommendation rewards against users whose true preferences are latent. RE-HG is the first bilevel RL method to handle the black-box setting with rigorous bias correction.

Why Existing Approaches Fall Short

White-box bilevel RL requires R_F explicitly. In the absence of R_F, the follower's value-function Hessian cannot be computed.

Naive IRL + hypergradient: Using an IRL-estimated $\hat{R}_F$ in place of R_F introduces reward-shaping bias. Any function of the form $\hat{R}_F + \gamma\Phi(s') - \Phi(s)$ fits the follower's trajectories equally well for any potential Φ, yet the resulting hypergradients differ. The gradient estimate is biased by an amount proportional to $\nabla_{\theta_L}\Phi$, which can be large.

Model-free bilevel RL avoids IRL but requires differentiating through the follower's policy directly, which is only feasible when the follower's policy is a differentiable network under the leader's control.

Core Mechanism

failure modes of naive bilevel RL with an estimated reward:

(1) Inverse RL reward estimation. MaxEnt IRL with function approximation recovers $\hat{R}_F \approx R_F$ from follower trajectories. The IRL estimate is updated online and tracks changes in $\pi_F^*(\theta_L)$.

(2) Bias cancellation via dynamic diversity. RE-HG collects follower trajectories under K diverse environmental dynamics and averages the resulting IRL rewards:

$$\bar{R}_F = (1/K) \sum_{k=1}^{K} \hat{R}_F^k$$

Under mild conditions, the shaping term satisfies $\sum_k \nabla_{\theta_L} \Phi^k \approx 0$ when the dynamics are sufficiently diverse.

(3) Eigenvalue truncation. The hypergradient involves a matrix inverse $(H_F)^{-1}$. RE-HG replaces it with $(H_F + \varepsilon I)^{-1}$ where ε is chosen to bound the inverse norm.

Technical Formulation

The leader's hypergradient via the implicit function theorem:

$$\nabla_{\theta_L} V_L = \partial_{\theta_L} V_L - (d\pi_F^*/d\theta_L)^T \nabla_{\pi_F} V_L$$

RE-HG substitutes $\bar{R}_F$ for R_F and replaces the Hessian inverse with the truncated version:

$$\hat{H}_F^{-1} = (H_F(\hat{R}_F) + \varepsilon I)^{-1}$$

The resulting RE-HG hypergradient estimate is then applied in gradient ascent on V_L.

Learning or Inference Procedure

RE-HG alternates between: (1) Follower simulation: roll out follower trajectories under $K = 5$ diverse dynamics for the current θ_L; update $\hat{R}_F$; (2) Hypergradient estimation: compute RE-HG using the averaged estimated reward and truncated Hessian; (3) Leader update: apply gradient ascent on V_L using the RE-HG estimate. No additional models are required beyond the follower policy, the leader policy, and a reward function approximator for IRL.

What the Guarantee Says

trajectories from K sufficiently diverse dynamics are collected, the RE-HG gradient estimate converges to the true hypergradient as $K \to \infty$, with bias decreasing as $O(1/K)$ times the shaping gradient's magnitude. No formal sample complexity bound is provided.

Experimental Findings

- **Gradient alignment:** Mean cosine similarity to white-box oracle is 0.91 across tasks; naive IRL baseline achieves 0.63.
- **Smooth landscapes:** Final V_L is statistically indistinguishable from oracle on 4/5 evaluated smooth-landscape tasks.
- **Sharp landscapes:** RE-HG outperforms oracle by 8% on average due to implicit Hessian regularization that prevents the oracle from getting stuck in saddle points.
- **Dynamics diversity K:** K = 3 recovers 85% of the bias cancellation; K = 5 achieves 97%; K = 10 adds negligible improvement.

Ablations and Interpretation

Without bias cancellation: Gradient cosine similarity drops from 0.91 to 0.74; on 2/5 tasks the leader converges to the wrong equilibrium.

Without eigenvalue truncation: Training is unstable in 3/5 tasks (Hessian norm exceeds 10^4); with truncation ($\varepsilon = 10^{-3}$), all 5 tasks converge stably.

IRL architecture choice: MaxEnt IRL with a 2-layer MLP reward function suffices; a 4-layer network improves alignment by <1%; a linear reward function loses 15% alignment.

Dynamic diversity design: Initial-state randomization contributes more to bias cancellation than transition perturbation; combining both achieves better cancellation than either alone.

Reference

NeurIPS 2026 Submission. **Reward-Estimated Hypergradient for Bilevel Reinforcement Learning with Black-Box Follower.** arXiv:2611.11998. NeurIPS 2026. arxiv.org/abs/2611.11998